HTML Helpers in ASP.NET Core MVC

🔷 What are HTML Helpers?

HTML Helpers are **methods used in Razor Views** to generate HTML elements dynamically.

They are part of:
👉 ASP.NET Core MVC

Instead of writing raw HTML manually, you use helper methods that:

- Bind data from the model
- Reduce repetitive code
- Support validation and model binding

🔷 Why do we use HTML Helpers?

Without HTML Helpers:

<input type="text" name="UserName" value="John" />

With HTML Helpers:

@Html.TextBox("UserName", "John")

Benefits:

- Strong integration with **Model**
- Automatic **value binding**
- Built-in **validation support**
- Cleaner Razor syntax

🔷 Types of HTML Helpers

There are **3 main types**:

1️⃣ Standard (Weakly Typed) HTML Helpers

These helpers use **string-based names**.

Example:

```
@Html.TextBox("UserName")

@Html.Password("Password")

@Html.TextArea("Address")
```

Key Point:

❗ No compile-time checking → error-prone

2️⃣ Strongly Typed HTML Helpers

These are the **most important in real projects**.

They use **lambda expressions** and are tied to the model.

Example Model:

```
public class Employee

{

   public string Name { get; set; }

   public int Age { get; set; }

}
```

In View:

```
@model Employee


@Html.TextBoxFor(m => m.Name)

@Html.EditorFor(m => m.Age)
```

Advantages:

✔ Compile-time safety
✔ IntelliSense support
✔ Refactoring-friendly

3️⃣ Templated HTML Helpers

Used to generate UI automatically based on data type.

Examples:

@Html.DisplayFor(m => m.Name)

@Html.EditorFor(m => m.Name)

👉 Automatically selects UI based on type

🔷 Commonly Used HTML Helpers

Let’s go category-wise:

🔹 Input Helpers

@Html.TextBoxFor(m => m.Name)

@Html.PasswordFor(m => m.Password)

@Html.HiddenFor(m => m.Id)

🔹 Label Helpers

@Html.LabelFor(m => m.Name)

Generates:

<label for="Name">Name</label>

🔹 Validation Helpers

@Html.ValidationMessageFor(m => m.Name)

@Html.ValidationSummary()

👉 Works with **Model Validation**

🔹 Display Helpers

```
@Html.DisplayFor(m => m.Name)
```

👉 Used in read-only views

🔹 Form Helpers

```
@using (Html.BeginForm())
{
   @Html.TextBoxFor(m => m.Name)
   <input type="submit" />
}
```

👉 Generates <form> tag

🔹 Dropdown & List Helpers

```
@Html.DropDownListFor(m => m.DepartmentId, Model.Departments)
```

🔷 HTML Attributes in Helpers

You can add HTML attributes easily:

```
@Html.TextBoxFor(m => m.Name, new { @class = "form-control", placeholder = "Enter name" })
```

🔷 How HTML Helpers Work Internally

- Use **Model Metadata**
- Generate HTML with correct name and id
- Enable **Model Binding** during POST
- Integrate with **Validation Framework**

🔷 HTML Helpers vs Tag Helpers

Feature	HTML Helpers	Tag Helpers
Syntax	C# method	HTML-like
Readability	Moderate	High
Usage	Older style	Modern approach

Example:

HTML Helper

```
@Html.TextBoxFor(m => m.Name)
```

Tag Helper

```
<input asp-for="Name" />
```

👉 Tag Helpers are preferred in modern ASP.NET Core

🔷 Real-Time Example (Complete Form)

```
@model Employee


@using (Html.BeginForm())

{

   <div>

       @Html.LabelFor(m => m.Name)

       @Html.TextBoxFor(m => m.Name)

       @Html.ValidationMessageFor(m => m.Name)

   </div>


   <div>
```

```
        @Html.LabelFor(m => m.Age)

        @Html.TextBoxFor(m => m.Age)

    </div>


    <button type="submit">Submit</button>

}
```

🔷 When to Use HTML Helpers?

✔ When working with **Razor Views**
✔ When you need **strong model binding**
✔ When maintaining **legacy MVC apps**

🔷 Summary

HTML Helpers:

- Simplify HTML generation
- Bind UI with Model
- Support validation
- Improve maintainability
